QE_ATF App Summary

What It Is

QE_ATF is a Streamlit-based automated testing framework for data migration and ETL validation. Repo evidence shows it supports source-to-target testing, data quality checks, SQL generation, profiling, and PDF/HTML reporting.

Who It's For

Primary persona: QA / data quality engineers validating data pipelines, migrations, and source-to-target mappings.

What It Does

- Create and manage test configs and protocol files from the UI.
- Run count, content, duplicate, null, schema, and fingerprint checks.
- Generate SQL from STTM/source-to-target mappings and bulk prompts.
- Profile source and target datasets and export HTML reports.
- Run data quality suites from JSON rules with PDF outputs.
- Store execution state in SQLite and generate summary/detail reports.

How To Run

- Install deps: pip install -r requirements.txt
- Start app: streamlit run datf_app/daqeai_portal.py
- Alternative packaged run exists in app.yml with the same Streamlit command.
- Docker helper scripts exist under datf_core/scripts/; production deployment details are Not found in repo.

How It Works

- UI: Streamlit entry at datf_app/daqeai_portal.py with feature pages in datf_app/pages/.
- App services: datf_app/common/commonmethods.py handles uploads, protocol loading, AI SQL prompts, profiling, and job dispatch.
- Execution layer: PySpark-based runners in datf_core/src/s2ttester.py, dqtester.py, and connections2t.py.
- Config/data: Excel protocols, STTM files, SQL, JSON connections, and sample data live under datf_core/test/.
- State/output: SQLite DBs in datf_core/utils/ plus generated HTML/PDF reports in datf_core/utils/reports and datf_core/test/results/.
- Runtime routing: testconfig.py switches execution between Docker and Databricks jobs.

Source basis: README.md, app.yml, testconfig.py, datf_app/, datf_core/scripts/, datf_core/src/, datf_core/test/, datf_core/utils/.